EXPERIMENT 4

Computer Networks — Star Topology using a Switch in Cisco Packet Tracer

Aim: To design a star topology with four PCs connected to a switch in Cisco Packet Tracer and verify connectivity between the hosts using the ping command.

Tools Used

Cisco Packet Tracer, 1 × 2960-24TT Switch (Switch0), 4 × PC-PT end devices (PC0 – PC3), Copper Straight-Through cables.

Procedure

1. Place a 2960 switch and four PCs on the logical workspace.
2. Connect each PC's FastEthernet port to the switch using copper straight-through cables.
3. Assign IP addresses in the 192.168.1.0/24 network to each PC (Desktop → IP Configuration).
4. Open the Command Prompt on PC0 and ping the other hosts (192.168.1.2, 192.168.1.3).
5. Switch to Simulation mode and observe the ICMP packet flow PC0 → Switch0 → PC2 → Switch0 in the Event List.

Output

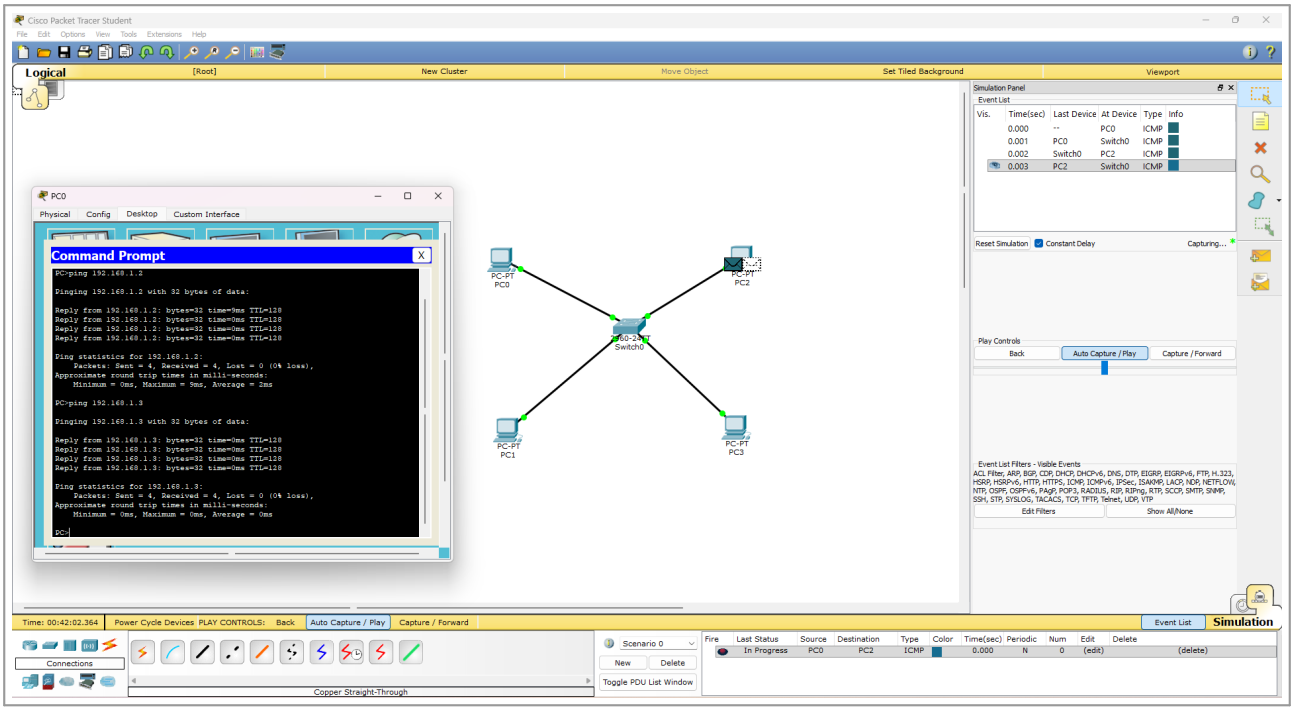


Fig 1: Star topology with successful ping from PC0 and ICMP events in Simulation mode

Observation

All 4 packets sent to 192.168.1.2 and 192.168.1.3 were received with 0% loss. The Event List shows the ICMP request travelling from PC0 through Switch0 to PC2 and the reply returning through the switch.

Result

The star topology was designed successfully in Cisco Packet Tracer, and connectivity between the PCs through the switch was verified using ping.